

Dampers and Intro to Laplace

EGR 345

GVSU

Outline

- quick review
- clean-up vertical mass spring EOM
- what is a damper?
- intro to Laplace

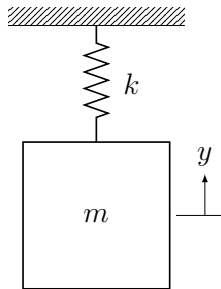
Quick Review

- what is this class about?

Quick Review

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- what is a dynamic system?
- what is feedback control?
- why are we focusing on mass/spring/damper systems?

Vertical MSD Clean-Up



- assume y is measured from the undeformed length of the spring
- define $\hat{y}$ as the distance from the point where gravity and the spring force cancel:

$$y = \hat{y} + y_{static}$$

What is a damper?

- what did you learn from the video?
- why do suspensions include a damper?
 - what happens to mass/spring systems that do not have a damper?
- how do we find the force of the damper mathematically?
 - $f = kx$ for a spring
 - $f = \gamma \dot{x}$? what a damper?

Damper Experiment

- what do we notice about the response of the system without a damper?
- what is different once the damper is included?

Intro to Laplace

Background and History of Laplace and Fourier Transforms

- Laplace History:

- `https:`

- `//en.wikipedia.org/wiki/Laplace_transform#History`

- Fourier History:

- `https:`

- `//en.wikipedia.org/wiki/Fourier_transform#History`

- **Key Ideas:**

- dudes in the 1800's played around with integrals for funsies
 - the internet hadn't been invented yet
 - some of their integrals turned out to be really helpful

Definition of the Laplace Transform

$$\mathcal{L}[f(t)] = \int_0^{\infty} f(t)e^{-st} dt$$

- unless you are an unbelievable mathematical genius, this shouldn't mean anything yet

How does the Laplace transform help us?

Key property of the Laplace transform:

- the Laplace transform turns differential equations into algebra

$$\frac{df}{dt} \rightarrow sF$$

(assuming zero initial conditions)

multiplying by s in the Laplace domain is the same as differentiating with respect to time

Foundational Laplace Transform

$$\mathcal{L} \left[\frac{df}{dt} \right] = sF(s) - f(0)$$

- let's derive this!
 - math is fun

Learning Activity 1

- review some diff. eq.

Laplace Analysis for LA 1

- probably won't have time for this